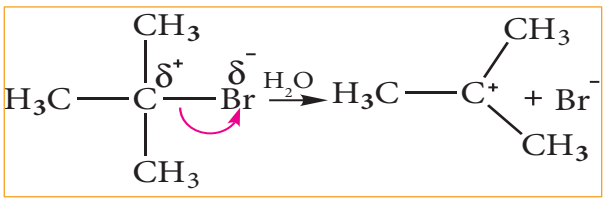
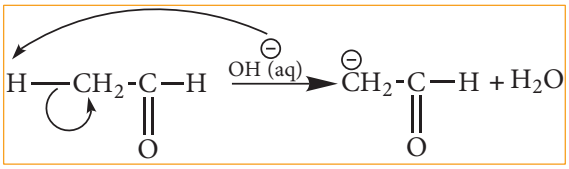
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For example, in tert-butyl bromide, the C-Br bond is polar as bromine is more electronegative than carbon. The bonding electrons of the C-Br bond are attracted more by bromine than carbon. Hence, the C-Br undergoes heterolytic cleavage to form a tert-butyl cation during hydrolysis.




Let us consider the cleavage in a carbon-hydrogen (C-H) bond of aldehydes or ketones We know that the carbon is more electronegative than hydrogen and hence the heterolytic cleavage of C-H bonds results in the formation of carbanion (carbon bears a negative charge). For example in aldol condensation the $OH^-$ ion abstracts a α-hydrogen from the aldehyde, which leads to the formation of the below mentioned carbanion.




## Hybridisation of carbon in carbocation:

In a carbocation, the carbon bearing positive charge is $sp^2$ hybridised and hence it has a planar structure. In the reaction involving such a carbocation, the attack of a negatively charged species (nucleophiles) take place on either side of the carbocation as shown below.

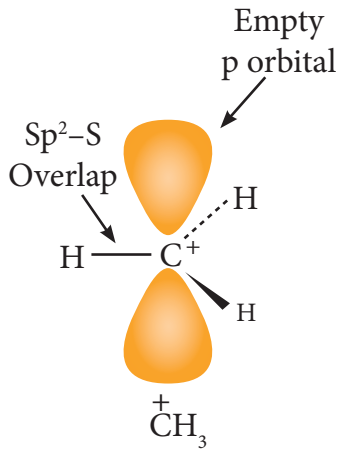



**Fig 12.1 Shape of Carbocation, Carbanion and free radicals**

The carbanions are generally pyramidal in shape and the lone pair occupies one of the $sp^3$ hybridised orbitals. An alkyl free radical may be either pyramidal or planar.

